COMP3242 Week 2 Practice Exam (Questions)

Instructions

Answer all questions. Show derivation steps where needed.

Q1: Logistic Regression (Calculation)

For $x = [1, 2]^T$, $a = [0.5, -1]^T$, and $b = 0.2$:

(a) Compute $z = a^T x + b$.

(b) Compute $\sigma(z) = \frac{1}{1+e^{-z}}$.

(c) If the true label is $y = 1$, compute binary cross-entropy loss $L = -\log(\sigma(z))$.

Q2: XOR and Expressivity (Conceptual)

Explain why a single linear classifier cannot solve XOR. Then explain how a two-layer perceptron can solve it at a high level.

Q3: MLP Parameter Count (Calculation)

A two-layer MLP has input dimension $n = 5$, hidden dimension $m = 8$, and output dimension $k = 3$. The model is

$$z = \phi(Ax + b), \quad y = Cz + d,$$

with $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times m}$, $d \in \mathbb{R}^k$. Compute the total number of trainable parameters.

Q4: Softmax vs Sigmoid (Concept + Application)

State one appropriate use case for each:

(a) Softmax + CrossEntropyLoss

(b) Sigmoid + BinaryCrossEntropy (or BCEWithLogitsLoss)

Also explain in one sentence why they are different.

Q5: Backprop Shape Tracing (Code/Debug)

In PyTorch, suppose:

```
x:      (64, 20)
A:      (50, 20)
b:      (50,)
C:      (10, 50)
d:      (10,)
```

For forward pass:

```
z = relu(x @ A.T + b)
logits = z @ C.T + d
```

Give shapes of ‘z‘ and ‘logits‘. Then state the expected label shape for ‘CrossEntropyLoss‘.